



## VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

# Workspace ONE Access Analytics / Audit Storage Down

WSA reports allOk=false and the health check FAILs — yet logins work. The OpenSearch analytics/audit store is stopped after a boot-time race with IDM. Root cause and the safe restart.

**v1.1**

Workspace ONE Access 3.3.7

# Workspace ONE Access — Analytics/Audit Storage Down (OpenSearch failed on boot)

**Prepared by:** Virtual Control LLC **Date:** July 13, 2026 **Document Version:** 1.1 **Product baseline:** Workspace ONE Access (vIDM) 3.3.7.0 (Build 21173100), Photon OS appliance **Classification:** Internal

**One-paragraph summary.** Workspace ONE Access (WSA / vIDM) reports `allOk=false` on its aggregate health endpoint — and a monitoring tool (e.g. the VCF Health Check) FAILs on it — **even though authentication and logins work normally.** The cause is that the **OpenSearch** service (WSA's analytics/audit storage) is **stopped**: it timed out "*Waiting for IDM*" during an earlier appliance reboot (a boot-time ordering race) and never started, so the audit/analytics service reports its storage as "*Connection refused*." Because OpenSearch backs only **audit/analytics** — not the auth path — logins are unaffected. The fix is a plain `systemctl restart opensearch` once IDM is up; it does **not** disrupt authentication. This can **recur on any appliance reboot** until a boot-ordering/timeout adjustment is made.

## 1.0 How to run the commands in this runbook

- **API health checks** run from **any host that can reach** `idm.lab.local` (a workstation, or on the appliance). They are unauthenticated `GET`s over HTTPS — copy the whole `curl` line.
- **Appliance commands** run **on the WSA appliance** over SSH as the `sshuser` account (member of `wheel`, so `sudo` -capable), then `sudo` for privileged actions. Replace `idm.lab.local` with your WSA FQDN.
- Commands are shown for a POSIX shell ( `bash` ). Where a command wraps across lines here, it is one logical command — copy the whole grey box.
- **OpenSearch** listens only on the appliance loopback (ports **9200** HTTP / **9300** transport); query it **from the appliance**, not remotely.

## 2.0 Symptom and Impact

- **Observed:** WSA aggregate health `GET https://<wsa>/SAAS/jersey/manager/api/health` returns `allOk: false`. Monitoring that keys on this flag raises a failure (e.g. VCF Health Check: "*Workspace ONE Access: health reports allOk=false*").
- **Misleading signal:** the *other* WSA health endpoint, `GET /SAAS/API/1.0/REST/system/health` (the **appmanager** view), reports `AllOk: true` with every sub-service (Database, Messaging, Analytics-connection, Clock, Encryption, Federation) `true`. **The two endpoints legitimately disagree** — see §3.
- **Impact is limited to audit/analytics:** OpenSearch stores WSA **audit events and analytics/reporting** data. **Authentication, SSO, directory sync, and logins are unaffected** (those are served by `horizon-workspace`,

which is up). Audit logging and the Analytics dashboards are degraded until OpenSearch is restored.

- **Trigger:** an **appliance reboot** (or a period of instability) in which OpenSearch's start script waited for IDM longer than its systemd start timeout and was killed.

### 3.0 Root-Cause Architecture

The aggregate `a110k` is a roll-up of four independent sub-systems — and only one is down. `GET /SAAS/jersey/manager/api/health` returns just `a110k` plus hypermedia `_links` to four sub-health endpoints:

| Sub-system               | Sub-health endpoint                                       | State in this incident                                                                                            |
|--------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| appmanager               | <code>/SAAS/API/1.0/REST/system/health</code>             | <b>healthy</b> ( <code>A110k=true</code> )                                                                        |
| messaging                | <code>/SAAS/jersey/manager/api/messaging/health</code>    | <b>healthy</b> ( <code>Connected</code> , 19/19)                                                                  |
| calculators              | <code>/SAAS/API/1.0/REST/system/health/calculators</code> | (auth-gated; not the culprit)                                                                                     |
| <b>analytics / audit</b> | <code>/AUDIT/API/1.0/REST/system/health</code>            | <b>FAILED</b> → <code>StorageConnectionOk=false</code> , <code>StorageStatus="Unknown: Connection refused"</code> |

The **analytics/audit** sub-system is the one dragging `a110k` to false. Its "storage" is **OpenSearch**; "*Connection refused*" means **nothing is listening** on the OpenSearch port — the service is down.

**Why OpenSearch was down.** The service unit is a System-V init script wrapped by systemd: `/etc/rc.d/init.d/opensearch` , with a drop-in `override.conf` setting `TimeoutSec=13min` . Its `start()` routine:

1. returns early if the OpenSearch daemon is **already running**;
2. checks whether `horizon-workspace` **is running** and starts it **only if it is not** (so it does *not* bounce an already-running auth stack);
3. prints `Waiting for IDM: ...` and blocks until IDM is reachable;
4. then launches the OpenSearch daemon ( `/opt/vmware/opensearch/bin/opensearch` , user `opensearch` ).

On the incident boots, **IDM was not ready within the 13-minute window** (fresh boot / instability), so step 3 never completed, systemd killed the start at the timeout ( `Active: failed (Result: timeout)` ), and the daemon at step 4 **never launched**. OpenSearch therefore stayed down across subsequent days — and the audit service kept reporting "*Connection refused.*"

### 4.0 Diagnostic Procedure (reproducible)

#### 4.1 Confirm which sub-system is failing (API, from anywhere)

```

curl -sk https://idm.lab.local/SAAS/jersey/manager/api/health      # aggregate: allOk
curl -sk https://idm.lab.local/SAAS/API/1.0/REST/system/health    # appmanager: AllOk +
DB/Messaging/Encryption/...
curl -sk https://idm.lab.local/AUDIT/API/1.0/REST/system/health    # analytics/audit:
StorageConnectionOk + StorageStatus

```

If `allOk=false` while `appmanager AllOk=true`, and the AUDIT endpoint shows `StorageConnectionOk=false` / **"Connection refused"**, the fault is the analytics storage (OpenSearch).

## 4.2 Confirm OpenSearch is down (appliance)

```

ssh sshuser@idm.lab.local
sudo systemctl list-unit-files --type=service | grep -iE 'opensearch|horizon' # -> opensearch.service,
horizon-workspace.service
sudo systemctl is-active opensearch                                           # -> failed (or inactive)
sudo systemctl status opensearch --no-pager | head -18                       # -> Active: failed
(Result: timeout); "Waiting for IDM: ..."
ss -ltn | grep -E ':9200|:9300'                                              # -> (nothing) =
OpenSearch not listening

```

## 4.3 Rule out the common false causes (appliance)

```

df -h | grep -vE 'tmpfs|devtmpfs'      # disk NOT full (a classic WSA OpenSearch cause) – all partitions
healthy here
free -m | head -2                      # memory NOT exhausted
sudo journalctl -u opensearch --no-pager -n 30 | tail -30 # shows the repeated "Waiting for IDM" ->
timeout pattern

```

**Confirm it is safe to restart before you do.** Read the init script's `start()` logic to verify a restart will **not** bounce authentication:

```

sudo grep -nE "horizon-workspace|Waiting for IDM|already running|serviceStatus"
/etc/rc.d/init.d/opensearch | head
cat /etc/systemd/system/opensearch.service.d/override.conf      # TimeoutSec=13min

```

The `start()` routine starts `horizon-workspace` **only if it is not already running**. When `horizon-workspace` is active (logins working), the restart leaves it untouched and only launches OpenSearch.

## 5.0 The Fix

**Restart OpenSearch once IDM is up (it is, if logins work). This does not disrupt authentication.**

```
sudo systemctl restart opensearch
```

Because IDM is already reachable, the "Waiting for IDM" step passes in seconds and the daemon launches. The command blocks until the init script completes (well under the 13-minute timeout in normal conditions). Then confirm the service and its ports:

```
sudo systemctl is-active opensearch          # -> active
ss -ltn | grep -E ':9200|:9300'              # -> LISTEN on 9200 and 9300
```

## 6.0 Recovery Behaviour — expect RED → YELLOW

Immediately after start, OpenSearch **recovers its shards** and reports `status: red` briefly (many shards still `initializing` / `unassigned`). This is normal. Watch it settle (on the appliance):

```
curl -s http://localhost:9200/_cluster/health?pretty
```

**YELLOW is the correct healthy state for this single-node cluster — not GREEN.** With `number_of_nodes: 1`, replica shards have nowhere to go, so they remain `unassigned` and the cluster caps at **YELLOW**. GREEN is unreachable single-node and is **not** required. The audit service treats YELLOW (primaries active) as healthy.

Once OpenSearch reaches YELLOW, the audit service reconnects (within a poll cycle) and the health flips:

```
curl -sk https://idm.lab.local/AUDIT/API/1.0/REST/system/health    # StorageConnectionOk -> true
curl -sk https://idm.lab.local/SAAS/jersey/manager/api/health      # allOk -> true
```

In this incident: OpenSearch went `red 48%` → `yellow 50%` (primaries recovered, 105 replicas unassigned = normal single-node), `StorageConnectionOk` → **true**, aggregate `allOk` → **true**. Total time a few minutes.

## 7.0 Recurrence and the Durable Fix (applied 2026-07-13)

**Without a change, this recurs on any WSA appliance reboot** that races OpenSearch's "Waiting for IDM" past the start window. The immediate remedy each time is the \$5 restart once IDM is up.

### 7.1 Why a `TimeoutSec` bump alone is NOT enough (verified, not assumed)

There are **two** independent gates on the boot-time wait-for-IDM, and the shorter one wins:

1. **The init script's own loop** in `/etc/rc.d/init.d/opensearch` (verified by reading the script): `MAX_ATTEMPTS=80`  
`# 80 x 10s = 800s (~13 min) until curl -k --fail ... https://localhost/SAAS ; do if [ ${COUNT} -eq ${MAX_ATTEMPTS} ]; then echo " Max attempts reached."; exit 1; fi printf '.'; COUNT=$((COUNT+1)); sleep 10`  
`done` At 80 attempts the **script itself** **exit 1 s** — before systemd's timeout is even reached.
2. **systemd's start timeout** — `TimeoutSec=13min` in the drop-in `override.conf`.

Raising only #2 does nothing, because #1 fires first. **Both** must be raised.

## 7.2 The fix that was applied

**Both gates widened to ~33–35 minutes, with backups. OpenSearch stayed running — the change takes effect on the next boot only.**

| File                                                                | Change                                                                                                      | Backup                                     |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <code>/etc/systemd/system/opensearch.service.d/override.conf</code> | <code>TimeoutSec=13min</code> → <code>TimeoutSec=35min</code>                                               | <code>override.conf.vc-bak-20260713</code> |
| <code>/etc/rc.d/init.d/opensearch</code> (line 105)                 | <code>MAX_ATTEMPTS=80</code> → <code>MAX_ATTEMPTS=200</code><br>( <code>200 × 10s = 2000s ≈ 33 min</code> ) | <code>opensearch.vc-bak-20260713</code>    |

Exact commands (root on the appliance):

```
cp /etc/systemd/system/opensearch.service.d/override.conf
/etc/systemd/system/opensearch.service.d/override.conf.vc-bak-20260713
cp /etc/rc.d/init.d/opensearch /etc/rc.d/init.d/opensearch.vc-bak-20260713
sed -i 's/TimeoutSec=13min/TimeoutSec=35min/'
/etc/systemd/system/opensearch.service.d/override.conf
sed -i 's/MAX_ATTEMPTS=80/MAX_ATTEMPTS=200/' /etc/rc.d/init.d/opensearch
systemctl daemon-reload
```

Verification (all confirmed on 2026-07-13):

```
cat /etc/systemd/system/opensearch.service.d/override.conf # -> TimeoutSec=35min
grep -n MAX_ATTEMPTS= /etc/rc.d/init.d/opensearch # -> 105: MAX_ATTEMPTS=200
systemctl show opensearch -p TimeoutStartUSec # -> TimeoutStartUSec=35min
systemctl is-active opensearch # -> active (unaffected)
```

The 33-min script wait sits safely inside the 35-min systemd timeout, so a slow boot has time to reach IDM — versus the ~13 min that failed on 07-09.

## 7.3 Caveats (verified reasoning — stated, not guessed)

- **Not reboot-tested.** The change is applied and confirmed effective in the running configuration; it has **not** been validated against an actual slow boot (that would require rebooting the appliance). The ~33-min margin is a deliberate safety choice over the ~13-min failure point, not a measured boot time.

- **WSA upgrades may reset** `/etc/rc.d/init.d/opensearch` (a vendor file), reverting `MAX_ATTEMPTS` . **Re-apply after any WSA upgrade.** The systemd drop-in is more likely to persist but re-check it too.
- **Rollback:** restore the two `*.vc-bak-20260713` files, then `systemctl daemon-reload` .
- If a boot ever still races past ~33 min, the §5 manual `systemctl restart opensearch` (once IDM is up) remains the fallback.

## 8.0 Verified Reference Facts

| Item                                                        | Value                                                                                                                                                                             |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product                                                     | Workspace ONE Access (vIDM) <b>3.3.7.0 Build 21173100</b> , Photon OS appliance ( <code>idm.la</code> <code>b.local</code> )                                                      |
| Aggregate health (the <code>allOk</code> monitors key on)   | <code>GET /SAAS/jersey/manager/api/health</code>                                                                                                                                  |
| Appmanager health (the misleading <code>AllOk=true</code> ) | <code>GET /SAAS/API/1.0/REST/system/health</code>                                                                                                                                 |
| Analytics/audit health (the real culprit)                   | <code>GET /AUDIT/API/1.0/REST/system/health</code> → <code>StorageConnectionOk</code> , <code>StorageStatus</code>                                                                |
| Storage engine                                              | <b>OpenSearch</b> ( <code>cluster_name: horizon</code> ), <code>/opt/vmware/opensearch</code> , user <code>opensearch</code> , ports <b>9200</b> (HTTP) / <b>9300</b> (transport) |
| Service unit                                                | <code>opensearch.service</code> (LSB <code>/etc/rc.d/init.d/opensearch</code> ), override <code>TimeoutSec=13min</code>                                                           |
| Auth service (kept running)                                 | <code>horizon-workspace.service</code>                                                                                                                                            |
| Appliance access                                            | SSH <code>sshuser@&lt;wsa&gt;</code> (group <code>wheel</code> → <code>sudo</code> )                                                                                              |
| OpenSearch cluster health (single-node normal)              | <b>YELLOW</b> (replicas unassigned by design; GREEN not achievable / not required)                                                                                                |

## 9.0 Document Information

|                          |                                                                                 |
|--------------------------|---------------------------------------------------------------------------------|
| <b>Title</b>             | Workspace ONE Access — Analytics/Audit Storage Down (OpenSearch failed on boot) |
| <b>Author</b>            | Virtual Control LLC                                                             |
| <b>Issued</b>            | July 13, 2026                                                                   |
| <b>Version</b>           | 1.1                                                                             |
| <b>Classification</b>    | Internal                                                                        |
| <b>Product baseline</b>  | Workspace ONE Access (vIDM) 3.3.7.0, Photon OS appliance                        |
| <b>Related Documents</b> | VCF9 Operations Campaign 2026-07-13 to 2026-07-17 (Open Item 6)                 |

|              |              |
|--------------|--------------|
| Distribution | Lab notebook |
|--------------|--------------|

## 10.0 Revision History

| Version | Date       | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | 2026-07-13 | Initial issue. Root-caused a WSA <code>allOk=false</code> (health-check FAIL, logins unaffected) to the <b>OpenSearch analytics/audit store being stopped</b> — it timed out " <i>Waiting for IDM</i> " during earlier reboots (boot-time race vs the <code>TimeoutSec=13min</code> start timeout). Documented the four-sub-system health roll-up (why the two health endpoints disagree), the reproducible diagnostic path, the safe restart ( <code>systemctl restart opensearch</code> — does not bounce <code>horizon-workspace /auth</code> ), the RED→YELLOW single-node recovery behaviour, and the reboot-recurrence caveat with optional durable fixes. Verified end-to-end: <code>StorageConnectionOk</code> and aggregate <code>allOk</code> returned to <code>true</code> .                      |
| 1.1     | 2026-07-13 | <b>Applied the durable fix</b> (§7) and corrected the record: a <code>TimeoutSec</code> bump <b>alone is insufficient</b> — the init script has its own <code>MAX_ATTEMPTS=80</code> (~800s) wait loop that <code>exit 1</code> s first (verified by reading the script). Raised <b>both</b> gates — <code>override.conf</code> <code>TimeoutSec</code> <code>13min</code> → <b>35min</b> and the script <code>MAX_ATTEMPTS</code> <code>80</code> → <b>200</b> (~33 min) — with <code>.vc-bak-20260713</code> backups; <code>daemon-reload</code> ; verified effective ( <code>TimeoutStartUSec=35min</code> , <code>MAX_ATTEMPTS=200</code> , OpenSearch stayed <code>active</code> ). Documented the caveats: <b>not reboot-tested</b> , WSA upgrades may reset the vendor script, and the rollback path. |